S M E Y A T S K Y L A B S

SynapticBridge™

Autonomous Self-Correcting MCP Orchestration Platform

Product Requirements Document | v1.0
Smeyatsky Labs | Confidential
March 2026

Classification: Confidential — Smeyatsky Labs Internal

1. Executive Summary

SynapticBridge™ is an autonomous, self-correcting MCP (Model Context Protocol) orchestration platform built for enterprises deploying AI agents at scale. Unlike dumb-pipe MCP bridges that merely route tool calls, SynapticBridge introduces a Correction Learning Engine (CLE) that observes every human override, builds a pattern graph of corrective intent, and applies learned corrections predictively before dispatch — turning operator expertise into institutional intelligence.

The platform is architected around four principles: zero-trust secure execution, autonomous routing intelligence, governed observability, and a self-improving correction flywheel. It targets the fastest-growing segment of enterprise AI spend: agentic infrastructure that runs reliably, auditably, and without constant human babysitting.

Strategic Positioning

SynapticBridge fills a critical gap in the current AI toolchain: the space between raw LLM capability and enterprise-grade autonomous execution. It is a Smeyatsky Labs standalone commercial product, positioned for SaaS licensing with an open-core distribution strategy.

2. Problem Statement

The MCP ecosystem is growing rapidly, but the tooling around orchestration and security has not kept pace with the ambitions of enterprise adopters. Three failure modes are endemic:

2.1 Dumb Routing

Current MCP bridges match tool calls by name or description. They carry no memory of what went wrong, cannot reason about intent-to-tool mapping confidence, and collapse when agents attempt novel tool chains. Every miscall is a human interruption.

2.2 Non-Existent Security Model

Most bridges pass API credentials as ambient environment variables with no scoping, no TTL, and no per-call audit trail. A compromised agent context means a compromised execution environment. GDPR, SOC 2, and ISO 27001 requirements cannot be met by these architectures.

2.3 Zero Institutional Memory

When operators correct an agent's tool call, that correction evaporates. The next session starts from zero. Organisations building autonomous workflows are accumulating invisible technical debt: the same corrections are made repeatedly by different team members with no shared learning.

Market Signal

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> Post-March 2026 cloud sovereignty incidents (AWS UAE region disruption, GDPR enforcement actions across EU agentic deployments) have elevated enterprise appetite for auditable, governed AI execution layers. SynapticBridge addresses this directly.

3. Product Vision

SynapticBridge becomes the de-facto execution fabric for enterprise MCP deployments — the layer that makes agentic AI systems trustworthy, governable, and progressively smarter without continuous human supervision.

Mission: *Make autonomous AI execution as reliable and auditable as human-supervised execution — then exceed it.*

Target Outcomes by Year 2

- Reduction in agent-triggered human interruptions by 70% through CLE predictive correction
- Full zero-trust execution model certified against SOC 2 Type II and ISO 27001
- Pattern library of 500+ learned correction rules exportable across enterprise deployments
- Sub-5ms routing overhead at p99 on commodity cloud infrastructure

4. Target Market & ICP

Primary ICP — Enterprise AI Platform Teams

Segment	Mid-to-large enterprises (500+ employees) deploying AI agents in production
Verticals	Financial services, telecoms, healthcare IT, professional services
Buyer	Head of AI / VP Engineering / CISO
Pain Trigger	Failed agentic pilot, compliance audit flag, repeated agent miscalls in production
Deal Size	£40k–£250k ARR per deployment

Secondary ICP — AI-Native Startups

Segment	Series A–C startups building autonomous workflow products
Verticals	LegalTech, FinTech, DevOps automation, RPA replacement
Buyer	CTO / Head of Product
Pain Trigger	Scaling agent reliability beyond founder-level debugging
Deal Size	£8k–£40k ARR, usage-based pricing tier

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5. Architecture

SynapticBridge is a multi-layer platform. The core is a Rust-based execution runtime with a Python agent coordination layer, a DuckDB-embedded pattern store, and an OPA policy engine. All layers communicate over a typed internal event bus.

5.1 Secure Execution Fabric

Every tool call runs within a signed, scoped execution context. The Execution Fabric is responsible for credential scoping, TTL enforcement, and cryptographic audit logging.

Execution Context

- Each agent session is issued a short-lived execution token (JWT, 15-minute default TTL)
- Tool manifests declare capability contracts: read / write / execute / network — no ambient authority
- Credentials are injected per-call via SPIFFE/SPIRE workload identity — never stored in env
- Every call produces an immutable, signed audit event written to an append-only log

Tool Manifest Schema

Each registered tool declares its contract in a TOML manifest:

```
[tool.filesystem.read]
capabilities = ["read"]
scope = "workspace:current"
ttl_seconds = 900
network_egress = false
audit_level = "full"
```

5.2 Correction Learning Engine (CLE)

The CLE is the core differentiator. It captures the delta between what an agent called and what the operator corrected it to, stores these corrections in a typed pattern graph, and applies learned corrections predictively before dispatch.

Correction Capture

- Human override events are intercepted at the dispatch layer
- The CLE records: original intent, inferred context, called tool, corrected tool, correction metadata
- Corrections are annotated with operator identity for weighted trust scoring

Pattern Graph

- Corrections are clustered by intent vector similarity using lightweight embedding

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- Each cluster produces a correction rule: IF intent matches [pattern] AND context matches [context] THEN prefer [corrected_tool]
- Rules are confidence-scored and decay with time unless reinforced

Predictive Dispatch

- Before executing a tool call, the CLE scores it against the rule graph
- If a matching correction rule exists above the confidence threshold, the corrected tool is dispatched instead
- All CLE-modified dispatches are flagged in the audit log with the triggering rule ID

> **The Flywheel**
> Agent intent → Route prediction → CLE check → Corrected dispatch → Tool result → Human override (if any) → Correction pattern → CLE update → Better prediction. The system compounds operator expertise into platform intelligence with every correction.

5.3 Autonomous Routing Intelligence

The routing layer maps agent intent to tool calls with confidence scoring. It supports multi-hop tool chaining, dependency resolution, and circular call detection.

Intent-to-Tool Mapping

- Intent embeddings computed at the routing layer, matched against tool manifest embeddings
- Confidence score gates dispatch: high confidence routes automatically, low confidence escalates to operator
- Configurable confidence thresholds per tool category (e.g. filesystem writes require higher confidence than reads)

Multi-Hop Chaining

- Tool chains are declared as dependency graphs, not sequential lists
- Dependency resolver validates data flow compatibility before execution begins
- Circular chain detection runs at plan-time, not execution-time — fails fast with diagnostic output

Graceful Degradation

- If a tool is unavailable, the router attempts capability-equivalent substitution from the tool registry
- Degraded execution paths are logged with substitution rationale
- Hard failures (no substitute available) surface actionable operator alerts, not silent drops

5.4 Observability & Governance Layer

SynapticBridge ships with a first-class governance model. Policy-as-code is the control plane; real-time call graph visualisation is the observability plane.

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Policy-as-Code (OPA)

- Operators write Rego policies that are evaluated at dispatch time
- Example: DENY tool:filesystem WHERE context:public_agent AND capability:write
- Policies are versioned, tested with built-in policy unit test harness, and audited on change
- Policy violations are non-negotiable hard blocks — not warnings

Drift Detection

- Tool manifests declare expected behaviour signatures
- Runtime behaviour is compared against declared signatures on each call
- Statistical drift alerts fire when tool behaviour deviates beyond configured sigma threshold
- Drift events are escalated to SIEM via webhook or native connectors (Splunk, Datadog, GCP Cloud Logging)

Call Graph Visualisation

- Real-time DAG view of all active tool chains per session
- Historical playback of any session for post-incident review
- Correction overlays: show where the CLE intervened on any historical trace

6. Technology Stack

Layer	Technology	Rationale
Core runtime	Rust (Tokio async)	Zero-cost abstraction, memory safety, sub-ms latency
Agent coordination	Python 3.12 + uv	LLM ecosystem compatibility, rapid tool integration
CLE pattern store	DuckDB (embedded)	Zero-ops analytics, BigQuery sync for enterprise deployments
Policy engine	Open Policy Agent (OPA)	Industry standard, Rego expressiveness, audit-ready
Workload identity	SPIFFE / SPIRE	Zero-trust workload auth, no static credentials
Transport	STDIO + HTTP/SSE	Full MCP spec compliance, bi-directional streaming
Observability	OpenTelemetry + Prometheus	Vendor-neutral, SIEM-connectable
Cloud target	GCP (primary), AWS, Azure	GCP-first aligns with Searce partnership leverage

7. Security Architecture

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SynapticBridge is designed for enterprise security requirements from the first commit, not bolted on post-MVP. The security model is zero-trust throughout.

Threat Model Coverage

Threat Vector	Mitigation
Credential leakage via env vars	SPIFFE/SPIRE per-call injection; no persistent env credentials
Agent prompt injection → tool escalation	Capability contracts enforce hard ceiling; no runtime elevation
Lateral tool chain exploitation	Scoped execution tokens; cross-tool data flow type-checked
Audit log tampering	Append-only, cryptographically signed log; WORM storage option
Malicious tool manifest	Signed manifest registry; supply chain verification via SBOM
Policy bypass	OPA evaluated in isolated subprocess; policy violations are hard blocks

Compliance Targets

- SOC 2 Type II — audit log completeness, access control, availability
- ISO 27001 — information security management alignment
- GDPR Article 25 — privacy by design for any PII-handling tool chains
- UK Cyber Essentials Plus — baseline for UK public sector prospects

8. Product Tiers & Pricing

Tier	Community	Professional	Enterprise
Target	Individual / OSS	SMB / Startup	Mid-market / Enterprise
Deployment	Self-hosted	Cloud-hosted or self-hosted	Private cloud / on-prem
MCP Tools	Up to 10 tools	Up to 100 tools	Unlimited
CLE Rules	50 correction rules	Unlimited rules	Unlimited + rule export
Policies	3 OPA policies	Unlimited policies	Unlimited + policy templates
Audit log	30-day retention	1-year retention	Configurable / WORM
SLA	Community support	99.5% uptime SLA	99.9% + dedicated CSM
Pricing	Free / Open-core	From £800/month	From £3,500/month

Open-Core Strategy

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Community tier ships as open-source (Apache 2.0) to drive developer adoption and CLE pattern contribution. Professional and Enterprise tiers monetise governance, SLA, and multi-tenant capabilities. This mirrors the Hashicorp / Grafana playbook and accelerates ecosystem growth ahead of competition.

9. Competitive Landscape

Capability	Generic MCP Bridges	LangGraph Cloud	SynapticBridge™
Security model	API key passthrough	Platform-level RBAC	Zero-trust, SPIFFE/SPIRE
Correction learning	None	None	CLE — full flywheel
Governance / policy	None	Basic role gates	Policy-as-code (OPA)
Routing intelligence	Name matching	Graph-defined flow	Intent-aware + confidence
Compliance certs	None	SOC 2 (platform)	SOC 2 + ISO 27001 + GDPR
Audit trail	Basic logging	Run tracing	Cryptographic, immutable
Deployment model	Self-hosted	Cloud-only (US)	Self-hosted + cloud + air-gap
CLE rule portability	N/A	N/A	Exportable rule sets (org-to-org)

10. Phased Delivery Roadmap

Phase 1 — Foundation (Q2 2026, 12 weeks)

- Rust core bridge runtime: MCP spec compliant STDIO + HTTP/SSE transport
- Tool manifest schema, registry, and SPIFFE/SPIRE credential injection
- Basic CLE: correction capture, pattern storage in DuckDB
- OPA policy engine integration with 5 built-in policy templates
- Cryptographic audit log with local append-only store
- CLI tooling for tool registration, policy management, log inspection
- Open-core Community tier release on GitHub

Phase 2 — Intelligence (Q3 2026, 10 weeks)

- CLE predictive dispatch with configurable confidence thresholds
- Intent-to-tool embedding model (distilled, runs locally — no external API dependency)
- Multi-hop chain planner with dependency resolver and circular detection
- Real-time call graph visualisation (web dashboard)
- Drift detection with OpenTelemetry export
- Professional tier launch: cloud-hosted deployment, usage billing

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Phase 3 — Enterprise (Q4 2026, 10 weeks)

- CLE rule export / import protocol for org-to-org pattern sharing
- WORM audit log option with GCS / S3 backend
- SIEM connectors: Splunk, Datadog, GCP Cloud Logging, Azure Sentinel
- SOC 2 Type II audit preparation and certification engagement
- Air-gap deployment package (for sovereign cloud / financial services)
- Enterprise tier GA with dedicated CSM and SLA

Phase 4 — Ecosystem (Q1 2027)

- Public CLE pattern marketplace: organisations publish and consume correction rule sets
- Claude Code integration (native SynapticBridge MCP server for Claude Code sessions)
- Partner API for ISVs to embed SynapticBridge into their own agentic platforms
- Potential ClaudeGuard integration: SynapticBridge as execution layer under ClaudeGuard governance

11. Go-to-Market Strategy

Developer-Led Growth (Community → Professional)

Open-core release on GitHub with strong documentation, a public CLE demo, and Smeyatsky Labs thought leadership on LinkedIn and developer communities. Target Claude Code power users, LangGraph dissatisfied users, and enterprise architects searching for MCP governance tooling post-2026 cloud sovereignty events.

Enterprise Acceleration via Searce

Leverage existing Searce GCP pipeline for enterprise introductions. SynapticBridge positions naturally alongside GCP AI Platform and Vertex AI deployments as the agentic execution governance layer. Co-sell motion requires minimal modification to existing Searce account executive playbook.

Partnership Channels

- Joel / Exobrain — ClaudeGuard integration creates natural joint GTM for enterprise Claude deployments
- Amarnath / Greyquill — India delivery and distribution channel; SynapticBridge as part of Greyquill AI modernisation service packaging
- Google Cloud Marketplace listing — drives inbound from GCP-committed enterprises

Content & Positioning

- Publish post-incident RaaS and SynapticBridge joint positioning paper (sovereign cloud + agentic governance)
- Case study: Correction flywheel — X% reduction in agent interruptions in 30 days

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- LinkedIn series: "The Broken MCP Bridge" — 5-part thought leadership establishing the problem before announcing the solution

12. Financial Projections (Indicative)

Metric	Year 1 (2026)	Year 2 (2027)	Year 3 (2028)
Professional customers	15–25	60–80	150–200
Enterprise customers	3–5	15–20	40–60
ARR (Professional)	£144k–£240k	£576k–£768k	£1.44M–£1.92M
ARR (Enterprise)	£105k–£175k	£525k–£700k	£1.4M–£2.1M
Total ARR (conservative)	£249k	£1.1M	£2.84M
Headcount (engineering)	1–2 (Allan + 1)	3–4	6–8
Primary cost	Founder time + GCP	Engineering salaries	Engineering + GTM

Independence Milestone

At £500k+ ARR (projected mid-2027), SynapticBridge alone is sufficient to support full independence from Searce, aligning with the September 2026 vesting milestone and post-vesting transition plan.

13. Risks & Mitigations

Risk	Likelihood	Mitigation
Anthropic ships native MCP governance	Medium	Focus on CLE correction flywheel — this is a data and learning moat, not a features race
Open-source clone commoditises core bridge	High	Community tier is the hook; enterprise governance and CLE rule portability are the moat
Rust talent scarcity for scaling team	Medium	Founder-built Rust core; Python agent layer is hireable; Greyquill India resource for Python work
Enterprise sales cycle >6 months	High	Professional tier self-serve closes fast; enterprise pipeline runs in parallel via Searce
MCP spec volatility (Anthropic changes spec)	Low	Spec compliance layer abstracted; transport adapter pattern isolates spec changes to a single module

14. Success Metrics

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Product KPIs

- CLE correction accuracy: % of predictive corrections confirmed correct by operator — target >85% at 90-day mark
- Routing confidence accuracy: % of high-confidence routes that complete without override — target >95%
- Dispatch latency overhead: p99 added latency vs. direct tool call — target <5ms
- Security: zero credential exposure incidents in production deployments

Business KPIs

- GitHub stars: 500 within 90 days of open-core launch
- Community-to-Professional conversion: 8% within 6 months
- First paid Enterprise customer: within 120 days of Enterprise tier GA
- Net Revenue Retention: >120% by Year 2 (expansion via tool seat growth)

15. Appendix: Glossary

MCP	Model Context Protocol — Anthropic's open standard for connecting AI models to external tools and data sources
CLE	Correction Learning Engine — SynapticBridge's proprietary pattern learning system
SPIFFE/SPIRE	Secure Production Identity Framework for Everyone — CNCF workload identity standard
OPA	Open Policy Agent — CNCF policy-as-code engine using the Rego policy language
Zero-trust	Security model requiring explicit verification of every request regardless of network origin
Execution token	Short-lived, scoped JWT issued per agent session to govern tool access
Tool manifest	TOML schema declaring a tool's capability contract, scope, and audit requirements
Drift detection	Statistical monitoring of tool runtime behaviour against declared manifest signatures
WORM	Write Once Read Many — immutable storage pattern for audit logs
Air-gap	Fully isolated deployment with no external network access — for sovereign or classified environments

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